

PHY201 Practice Exam on chapter(s): 1, 2, 3

Name: _____

These questions assess your understanding of the material from Chapters 1, 2, 3 of OpenStax College Physics.

1. Victoria is skiing at a speed of 3.2 m/s just as she crests a hill. On the downslope, she accelerates at 5.2 m/s² for 4.2 s. Calculate her speed at the end of the 4.2 s.
ANSWER: _____
2. A man walks 16.9 m east and then 52.8 m south. What is the difference between his travelled distance and the absolute value of his displacement?
ANSWER: _____
3. Aaron throws a beanbag horizontally from his dorm window 15.3 m above the ground. Mark, another student, stands 7.52 m away from the dorm and catches the beanbag at a height of 1.62 m above the ground. Mark claims that Aaron threw the beanbag too slowly, so Aaron wants to calculate its initial speed to prove him wrong. Ignoring friction, what initial speed does he calculate?
ANSWER: _____
4. On a family trip, Thomas and Samantha take their daughter, Sarah, to visit the deepest man-made hole on Earth, the Kola Superdeep Borehole. Sarah shows little interest, however, and spends all her time on her cell phone. Thomas, fairly horrified at his daughter's lack of scientific interest, grabs the phone and throws it down the hole, which is 9 inches in diameter and 12,262 meters deep. Thomas manages to hear over Sarah's cries that it hit the bottom in 50.0 seconds. What was the velocity at which Thomas threw the cell phone?
ANSWER: _____
5. Natalie drops a coconut off a cliff and measures it striking the ground with a vertical impact velocity of 67.5 m/s. Determine the height of the cliff. Report your answer in feet, recalling that 3.28 ft = 1 m.
ANSWER: _____
6. Laika the moon dog jumps 6.59 m into the air. What is the vertical speed at which Laika leaves the ground? Laika is on the Moon and so her free-fall acceleration is 1.625 m/s². Note: Laika was a stray part-terrier mix from Moscow who, in 1957, became the first living creature to enter space and orbit Earth, clearing the celestial trail for human space exploration.
ANSWER: _____
7. Out of curiosity, you want to conduct some geographic measurements of your house and your friend Anna's. To get to Anna's, you have to walk 3.39 km west and then 3.13 km north. What is the compass direction of a line connecting your house to hers?
ANSWER: _____
8. If a marathon runner averages 10 mi/h, how long does it take her to run a 19.6-mi marathon?
ANSWER: _____
9. A 5.6-kg rock is shot up vertically at the same time that a 0.6-kg ball is projected horizontally. Neglecting friction, if both start from the same height, which hits the ground first? Do they hit the ground at the same time? Describe the motion in words.
ANSWER: _____
10. Hannah boards a light-rail commuter train each day for work. One day, Hannah decides to use her phone's stopwatch to measure the time it takes for the train to reach its top speed of 100.0 km/h from rest, and finds that it takes 29.4 s. What is the train's average acceleration?
ANSWER: _____
11. Stephen throws a 3.9-g coin straight downward from a 19-story window with a vertical speed of 11.2 m/s (approximately 25.1 mph). Ignoring the effect of the drag force, at what speed does the coin impact a patch of grass on the ground? Express the speed in meters per second. Take a story as roughly 10.3 feet. There are 3.28 feet in one meter.
ANSWER: _____

12. Nick is hanging from a pull-up bar inside an elevator that is accelerating downward at a rate of 7.2 m/s^2 . Nick's mass is 85.2 kg and the distance between the elevator floor and the bottom of his feet is 0.88 m. Nick lets go of the bar just as the elevator changes its acceleration to 2.1 m/s^2 downward. At what speed will he hit the elevator floor?
ANSWER: _____
13. The magnitude of a velocity vector is 10.5 m/s and its x component's magnitude is 4.4 m/s. What is the magnitude of its y component?
ANSWER: _____
14. Kyle is watching his dog, Zeus, play around in the backyard. Zeus is standing still until he sees a squirrel. Over 11.7 s, Zeus covers a distance later measured to be 49.3 m. Kyle is curious: how fast was Zeus moving at 11.7 s?
ANSWER: _____
15. Grace throws a brick horizontally from a bridge with an initial speed of 16.1 m/s. Find the angle at which the brick enters the water 4.2 seconds later. You may ignore air resistance.
ANSWER: _____
16. Justin is sitting in his folding chair on the beach, watching his friend Efe as he rides in a pontoon moving at a velocity of 2.84 m/s at an angle of 0.3° East of South. Efe then passes his sister, Mary, who is riding in a pontoon, which, from Efe's perspective, is travelling at a velocity of 3.98 m/s at an angle of 78.5° East of North. From Justin's perspective, what is the velocity of Mary on the pontoon?
ANSWER: _____
17. Eric kicks a soccer ball, initially at rest on the ground, with an initial velocity of 20.7 m/s at a launch angle of 56.8° above the horizontal. The ball flies through the air and lands back on the ground. Calculate its total flight time, assuming that air resistance is negligible.
ANSWER: _____
18. At the end of her foot race, Olivia decelerates from a velocity of 5.2 m/s until she comes to rest, taking 1.8 s to do so. How far does Olivia travel, measuring from the end of her race until she comes to rest?
ANSWER: _____
19. A flea just barely jumps over a 0.67-m-tall wall. How long is the flea in the air, measured from when it jumps to when it lands?
ANSWER: _____
20. To test the idea of surviving a free-falling elevator, Kevin conducts an experiment using a string-attached plank and a programmable toy frog. His makeshift elevator falls 5.2 m and the frog jumps 0.0004 s before impact. Previous tests show that these toys can endure impacts below 6.38 m/s. What is the minimum initial speed needed for the frog's vertical jump to make it impact the ground at a safe speed? (Note: The plank's mass is significantly greater than the toy's mass.)
ANSWER: _____
21. Victoria is walking on a bridge between two mountain peaks, when curiosity suddenly overcomes her and she throws a bottle straight down from the bridge to the distant vale below. If the bottle falls 17 m in 1.2 s, then how far will it have fallen in 12.4 s?
ANSWER: _____
22. A BASE jumper runs off a cliff with a (horizontal) speed of 3.4 m/s. Which of the following is closest to his overall speed after 1.99 seconds?
ANSWER: _____
23. Laika the moon dog jumps into the air to reach a height of 7.09 m. How long is Laika in the air? Laika is on the Moon and so her free-fall acceleration is 1.625 m/s^2 . Note: Laika was a stray part-terrier mix from Moscow who, in 1957, became the first living creature to enter space and orbit Earth, clearing the celestial trail for human space exploration.
ANSWER: _____
24. A cat covers a distance of 68.8 m in a time of 42.0 s. What is the average speed of the cat over this distance?
ANSWER: _____
25. Grace aims her arrow towards a target located 201.4 m away. The target's bull's eye aligns with the height from which Grace releases the arrow. The arrow is launched at an angle with an initial speed of 106.6 m/s. Find the angle at which Grace launched the arrow.
ANSWER: _____